Final Exam

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1 TASK 1: FIND A RECENT PUBLIC ARTIFACT ADDRESSING AI MISUSE

1.1 Public Artifact

- Title: "AI Hiring Tools Can Yield Racial Bias and Systemic Rejection”
- Released: May 26, 2026
- Link: https://hai.stanford.edu/news/ai-hiring-tools-can-yield-racial-bias-and-systemic-rejection

1.2 Application/Scenario/Domain of Misuse

Automated Candidate Resume Screening

1.3 Regulated Domain/Protected Class

- Regulated Domain: Employment
- Protected Classes Impacted: Race and Ethnicity

1.4 Evidence

https://news.stanford.edu/stories/2026/06/ai-hiring-tools-racial-bias-research

2 TASK 2: SUMMARY OF BIAS IN PUBLIC ARTIFACT

This report shows how automated hiring tools used by major companies are biased against job seekers with certain ethinicity. When filtering through applicants, these tools tend to **filter out Black and Asian candidates while favoring White candidates**. Instead of filtering based on their qualifications, the algorithm relies on older, biased data and clues like test games or educational background. As a result, qualified minority applicants are frequently rejected before a human recruiter ever sees their application.

Furthermore, the report shows how **broad summary statistics** can easily hide this discrimination and mislead readers. Looking at hiring numbers across an entire company might make the system seem fair, but breaking down it down job by job reveals clear violations of employment fairness laws. To make it worse, over 150 top companies use the same hiring software, thus a single flaw in the system gets repeated everywhere. For example,If a candidate is mistakenly rejected by this algorithm once, they will end up getting rejected as well from jobs across other companies as they are using the same tools.

3 TASK 3: DISCUSSION OF ARTIFACT/EVIDENCE METRICS DEMONSTRATING BIAS

3.1 Privileged/unprivileged groups

- Unprivileged Groups: Black and Asian applicants
- Privileged Group: White job applicants

3.2 Misleading graphs (Explain why a graph shown is misleading

The graph is showing one big overall average and hiding bias against black and asian applicants. For example, If the tooI hires a lot of Black candidates for entry-level warehouse jobs, but rejects them for high-paying manager jobs, the overall average makes it look like everything is fair. In order to fix this, graphs must break down the results job by job to reveal where bias is happening.

3.3 Source(s) of data bias

The AI was trained on past hiring history and old manager reviews and it picks up on individual information like colleges attended, resume wording, or soft skills that act as substitutes for race. As a result, the AI learns and repeats past human prejudice on a massive scale without having to know the candidates' race.

3.4 Source(s) of sampling bias

Because all over 150 huge Fortune 500 companies use the exact same AI hiring tool. This results in a Chain-Reaction where if it unfairly rejects a candidate at one company, that same system will automatically reject them at next companies. Candidates never get a fresh start.

3.5 Sampling methods used to collect data

The researchers conducted real world tracking by following real job seekers over time, tracking 3.4 million candidates making 4.0 million applications across 1,700 job postings.

3.6 Correlations found in the data

Chain-Reaction Rejections: Getting rejected by the AI at Company A wasn't just bad luck. It strongly increased the candidate's chance of being rejected by Company B—proving the system itself is biased across multiple employers.

3.7 Outcome measures: averages, standard deviations

- Averages: white 28%, black 19%, asian 22% (Recommended rate out of 100 people of each group)
- Standard Deviations: black - 0.12, asian - 0.09, white - 0.04

4 TASK 4: PROPOSE A METHOD FOR MITIGATING BIAS

4.1 Identified Issue

To address demographic bias within the hiring assessment tool, we target the disparity in initial interview selection rates between applicant groups. Historical hiring data contains selection ratios that fall below acceptable statistical fairness thresholds, causing the model to learn and perpetuate past hiring imbalances against protected classe

4.2 Data Inputs

Historical applicant feature records such as candidate skills, work experience, and protected attributes such as demographic group paired with hiring outcomes of 1 as shortlisted or 0 as not shortlisted.

4.3 Data Outputs

Re-weighted instance values assigned to each candidate record. These calculated weights adjust how much the model learns from each profile during training.

4.4 Method

We used reweighting strategy to balance the dataset during data preparation before the model is trained. Instead of altering candidate qualifications, reweighting adjusts the weights or importance given to different applicant profiles in the historical data. It calculates expected selection probabilities versus actual historical outcomes, assigning higher importance weights to underrepresented candidates who were hired and slightly lower weights to overrepresented combinations.

```
# Reweighting to reduce demographic bias in hiring
# Reference: https://arxiv.org/html/2312.12560v1

# Step 1: Start with historical hiring data.
# Each applicant has:
# - whether they were shortlisted
# - whether they are in the privileged demographic group

# Step 2: Measure how often applicants were shortlisted and rejected overall.

# Step 3: Measure how common the privileged group is and how common the unprivileged group is.

# Step 4: Break the data into four groups:
# 1. privileged + shortlisted
# 2. privileged + rejected
# 3. unprivileged + shortlisted
# 4. unprivileged + rejected
# Count how often each group appears in the data.

# Step 5: Compute a fair target size for each group
# Example: if 40% of applicants are privileged and 30% are shortlisted,
# then we'd expect about 40% * 30% = 12% to be privileged AND shortlisted
# Repeat for the other three groups.

# Step 6: For each group, compute a weight:
# weight = (fair target from Step 5) / (actual share from Step 4)
#
# Example:
# - Unprivileged + shortlisted is only 5% of the data, but fair target is 10%.
#   weight = 10% / 5% = 2
#   Meaning: when training, pretend each of these people appears twice
#   so the model pays more attention to them.
#
# - Privileged + shortlisted is 20% of the data, but fair target is 10%.
#   weight = 10% / 20% = 0.5
#   Meaning: when training, pretend each of these people appears only half
#   as often so the model is less swayed by that pattern.

# Step 7: Apply the weight to each applicant for each group.

# Step 8: Train the model using these weights
```

Figure 1—Pseudo Code for Reweighting with examples and comments

REFERENCES

[1] Blow, C. H., Qian, L., Gibson, C., Obiomon, P., & Dong, X. (2023). "Comprehensive validation on reweighting samples for bias mitigation via AIF360." *arXiv preprint arXiv:2312.12560*. URL: https://arxiv.org/abs/2312.12560.